

“ Benign prostatic hyperplasia, bladder outlet obstruction and urinary retention”



Division of Urology, Department of Surgery
Queen Mary Hospital
The University of Hong Kong

Urinary Retention

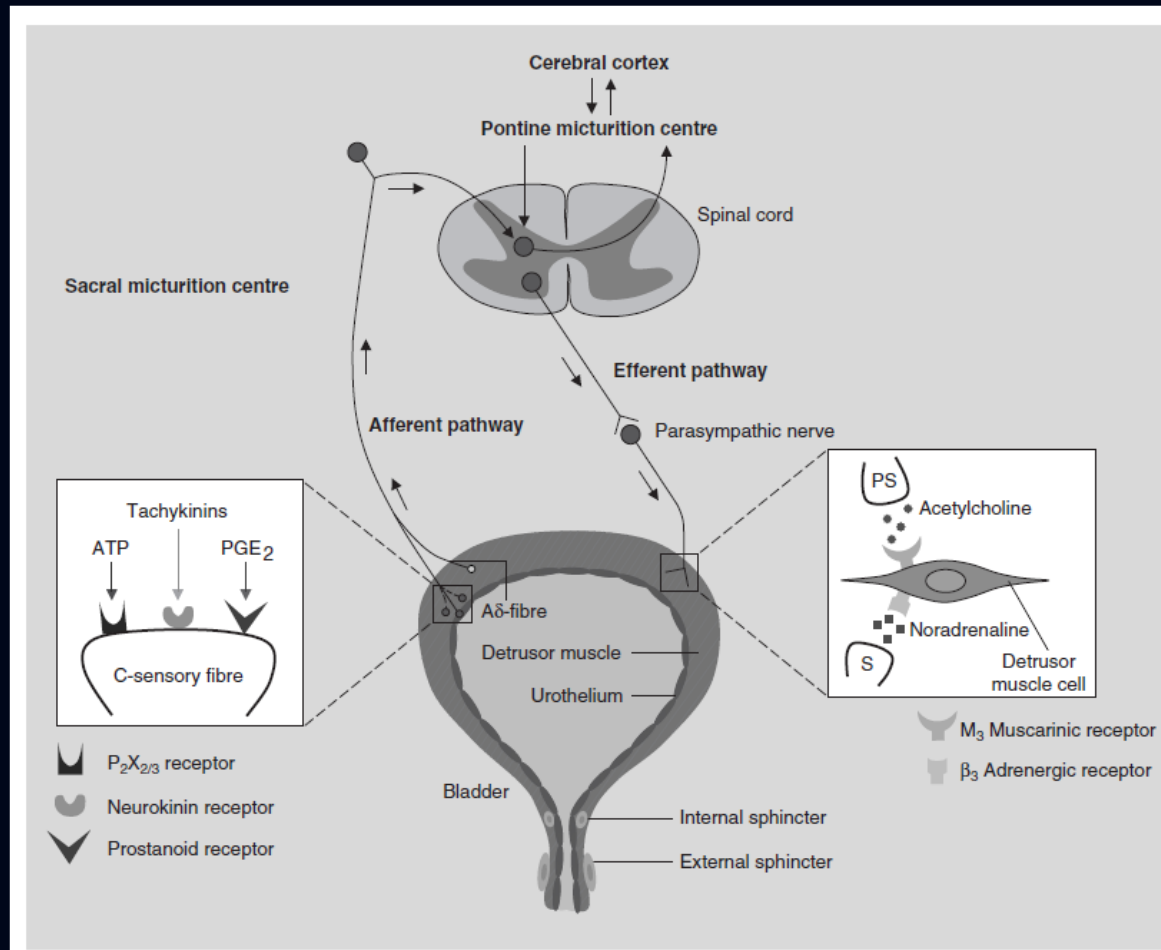


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Lecture

1. Physiology of micturition
2. Pathophysiology of urinary retention
3. Clinical management of urinary retention

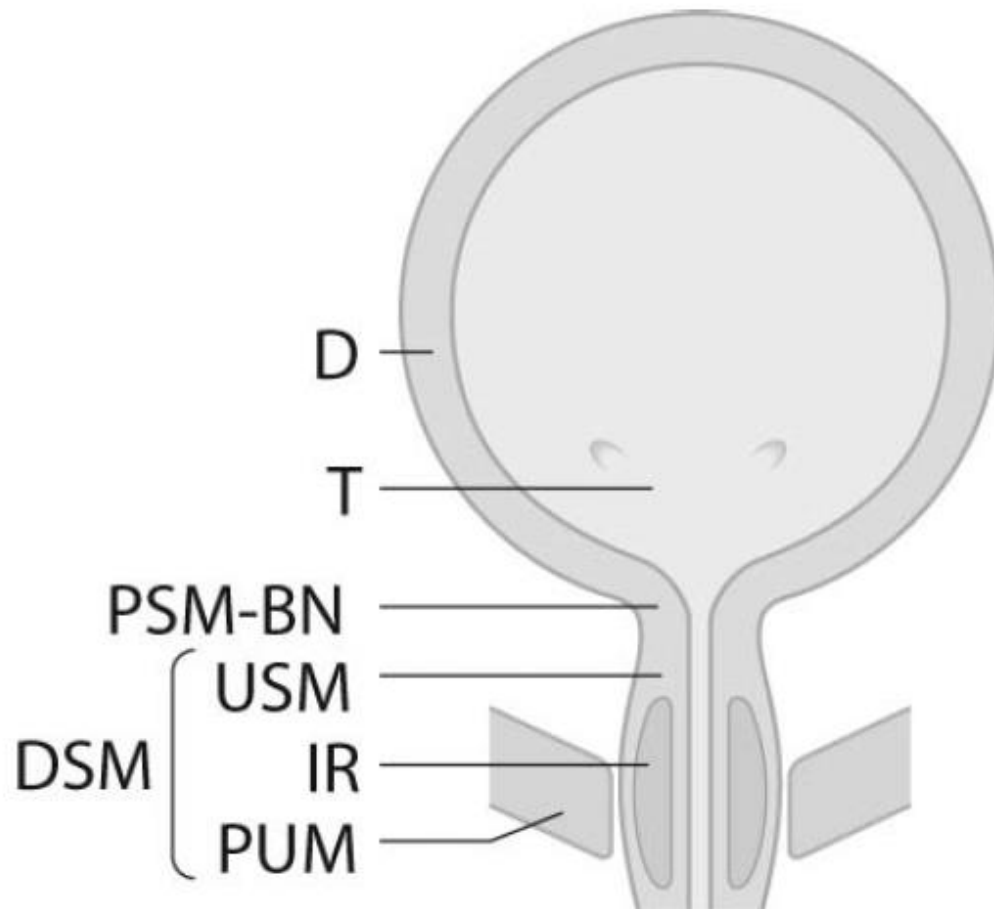
Physiology of micturition



Lower Urinary Tract

- Male
 - Urinary Bladder
 - Prostate
 - Urethra
- Female
 - Urinary Bladder
 - Urethra

Lower Urinary Tract



Normal voiding function

STORAGE

Accommodation of urine at low pressure and with appropriate sensation

Bladder outlet remain closed at rest and during increase intra-abdominal pressure

No involuntary bladder contraction

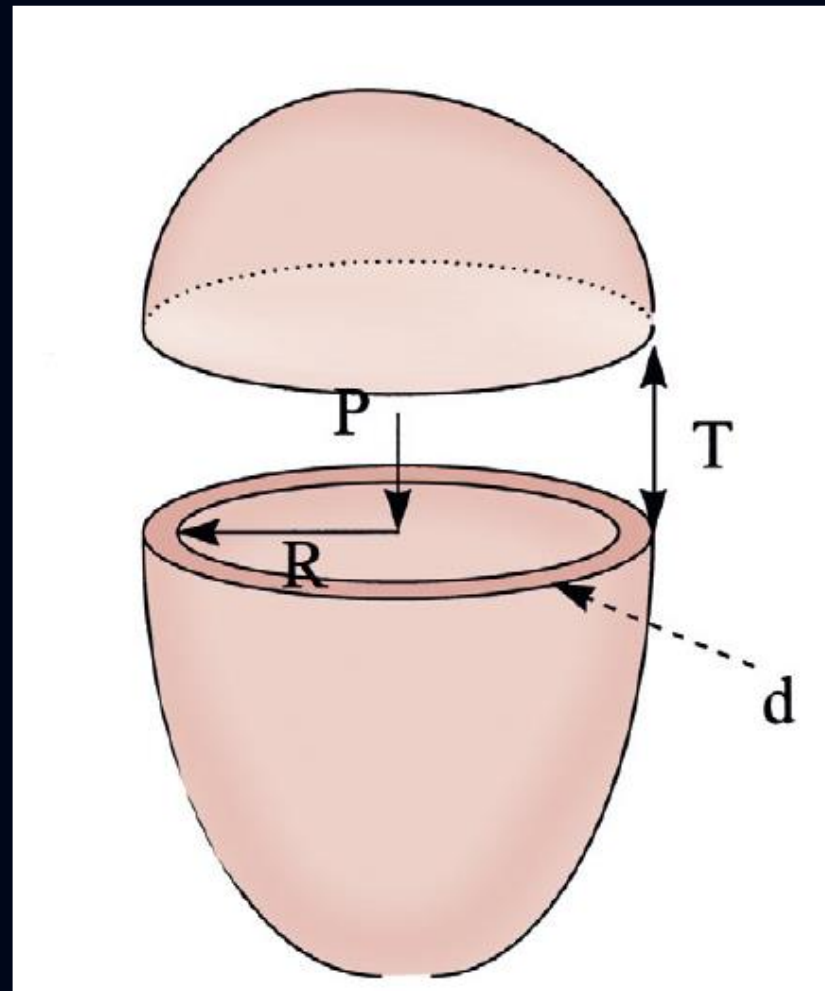
EMPTYING

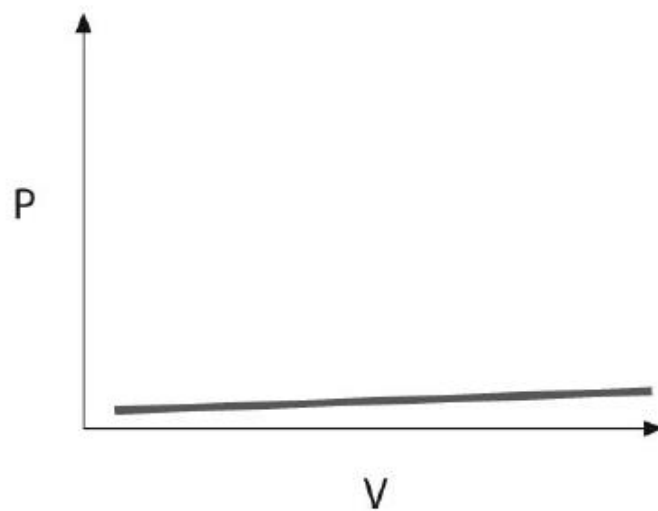
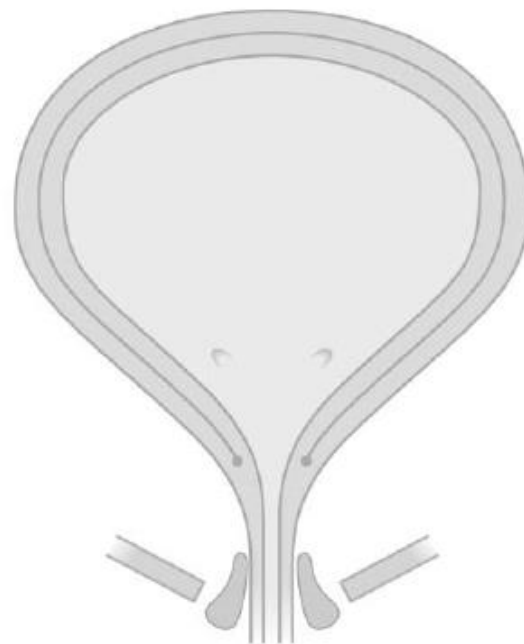
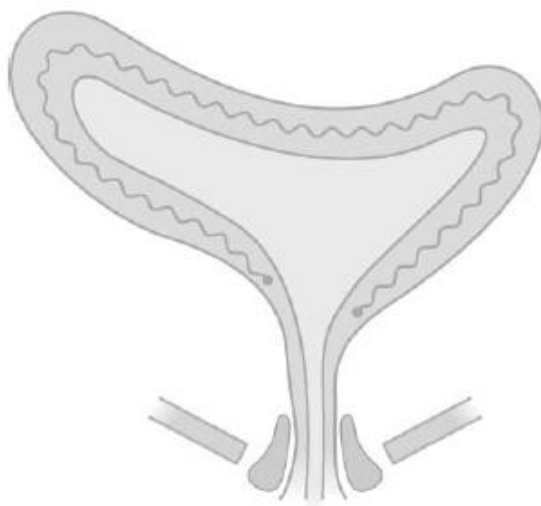
Coordinated contraction of bladder smooth muscle of adequate magnitude and duration

Lowering of resistance at the level of smooth and striated sphincter

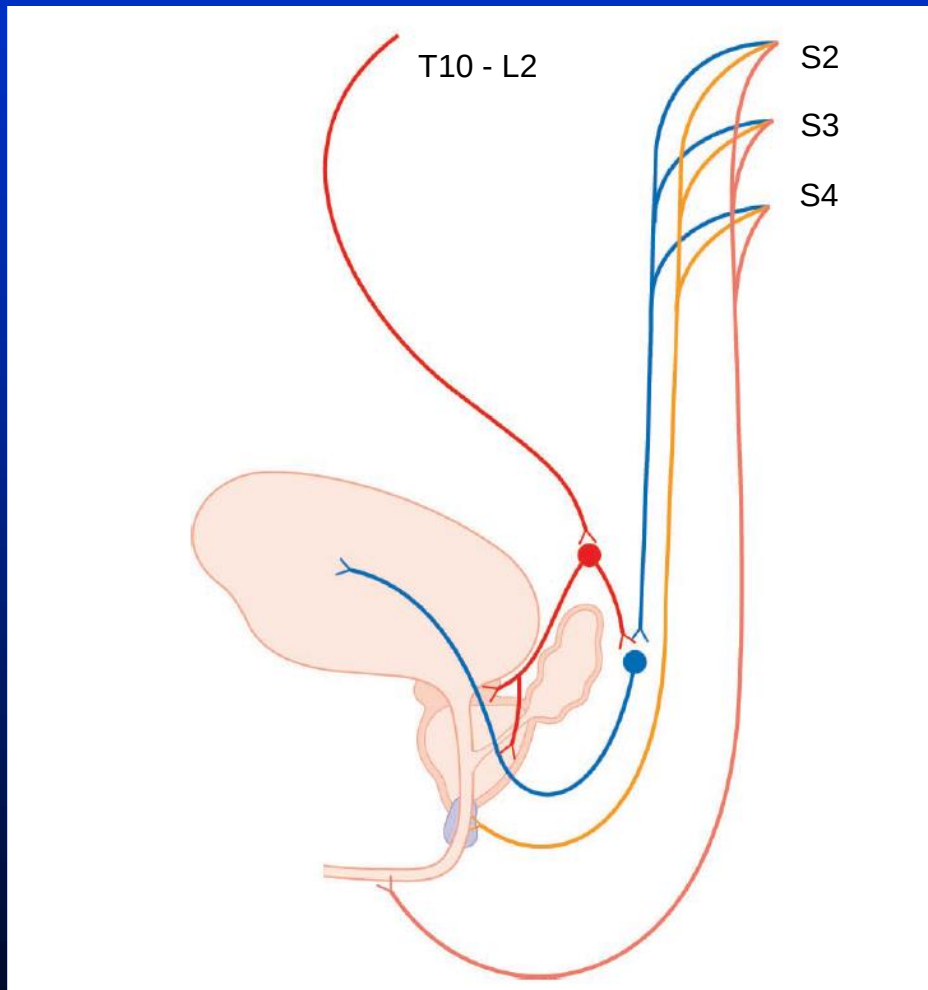
Absence of anatomic obstruction

$$T = P_{\text{ves}} R/2d$$





Efferent Innervation of the Lower Urinary Tract



Blue :
Parasympathetic

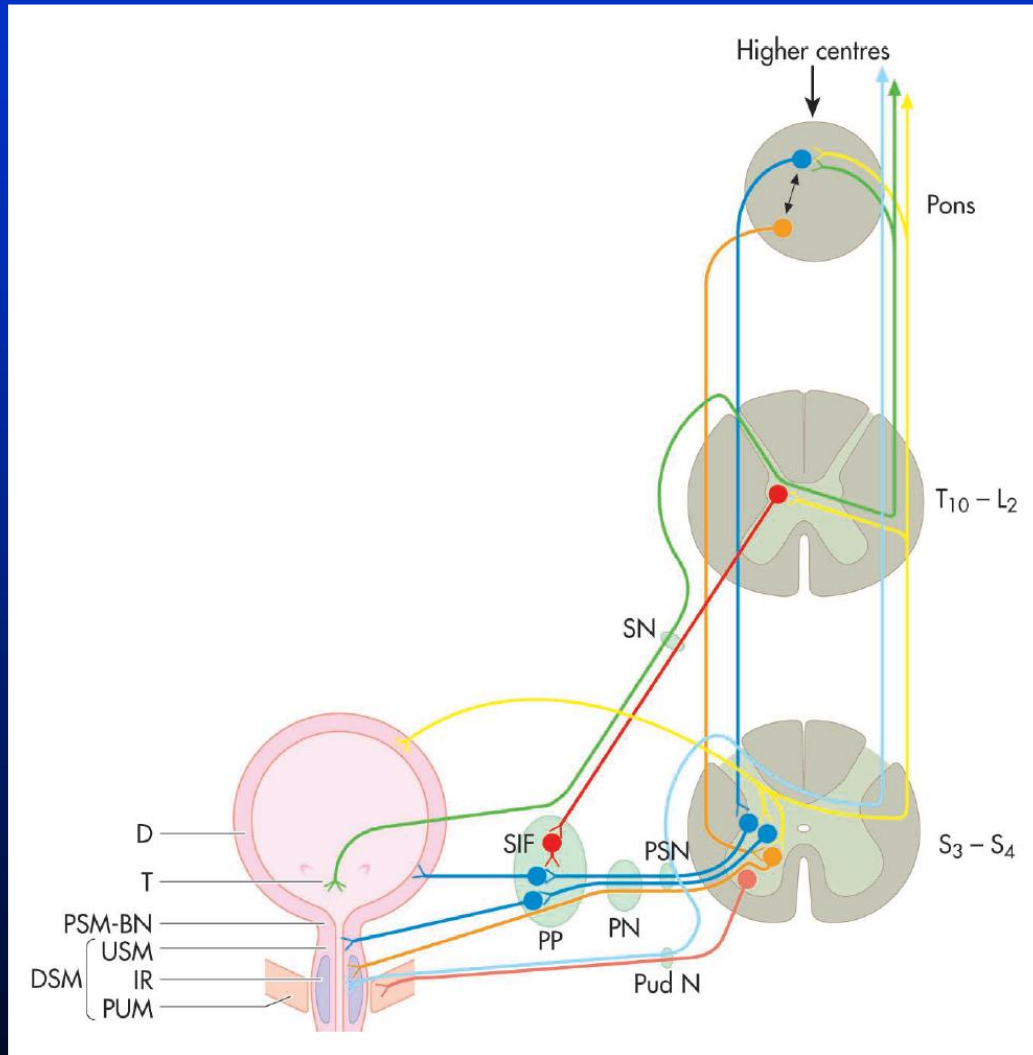
Red :
Sympathetic

Orange + Pink :
Pudendal/somatic

Efferent receptors

- Detrusor
 - β_3 adrenergic receptors : relaxation
 - M_3 muscarinic receptors : contraction
- Bladder neck / prostate
 - α_1 adrenergic receptors : contraction
- External / Rhabdosphincter
 - Nicotinic receptors : contraction

Afferent Innervation of the Lower Urinary Tract



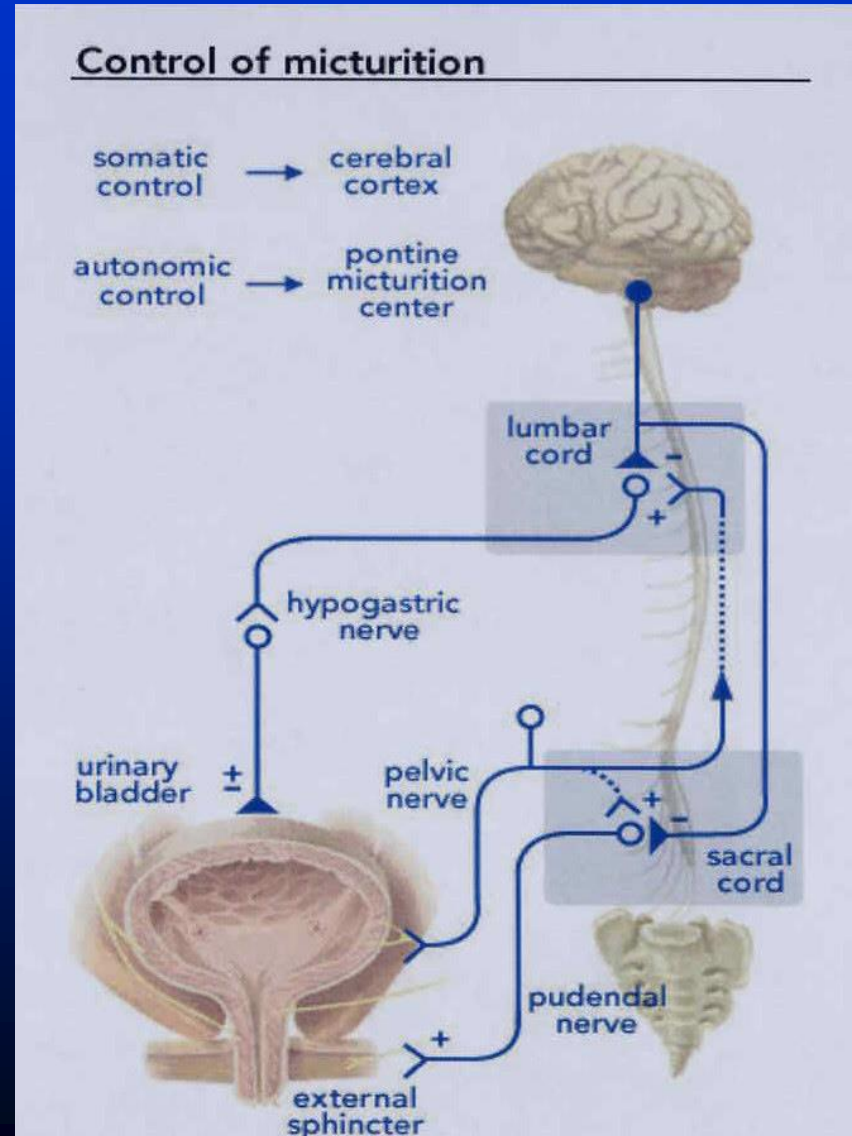
- Present in hypogastric, pelvic (majority) and pudendal nerves
- Myelinated A δ and unmyelinated C-fibres
- All relay to higher centers

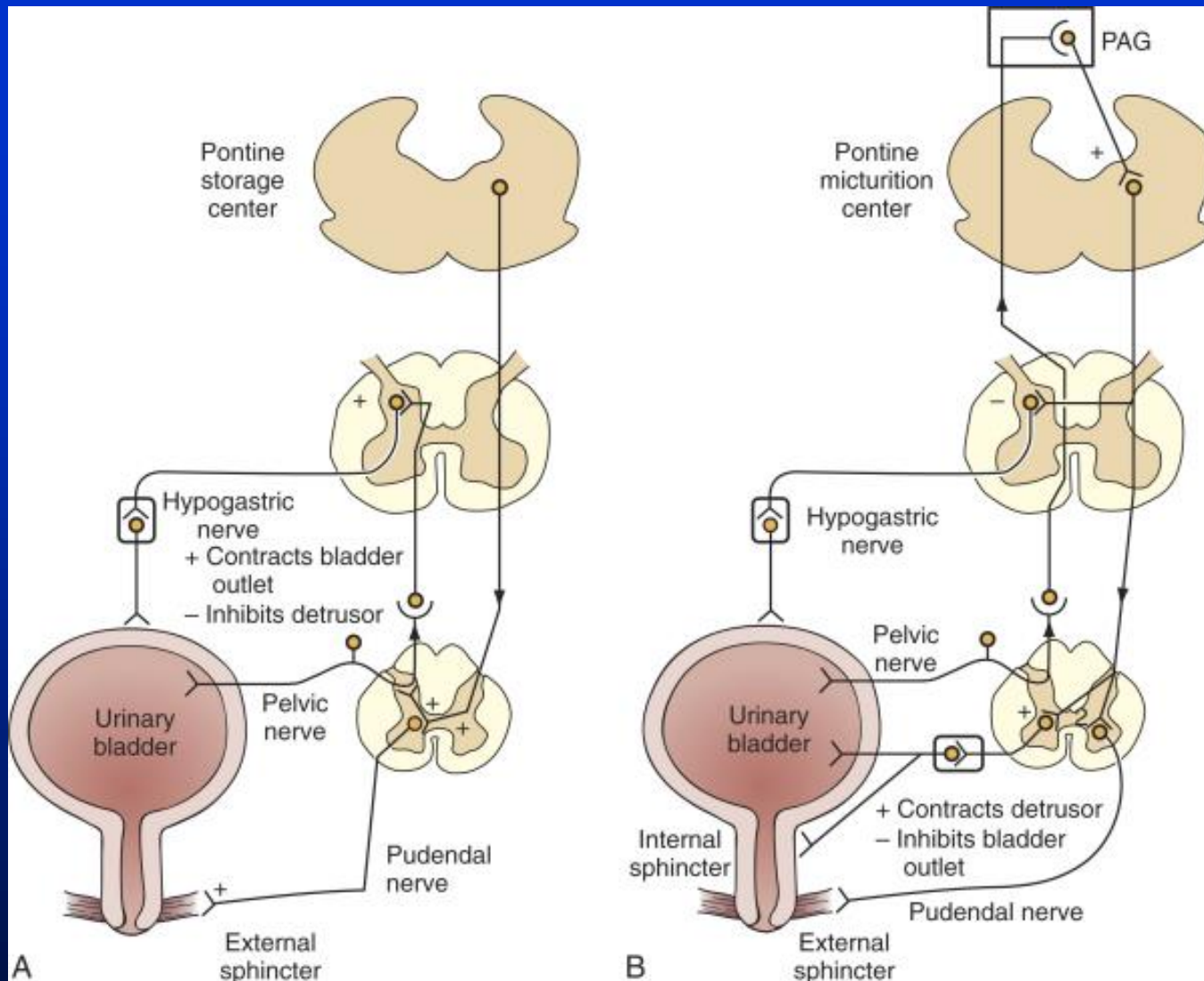
Brain control centers

- Pontine Micturition Centre (Barrington's nucleus)
- Suprapontine control (Cerebral cortex)

Innervation overview

- Sympathetic
 - T10-L2
 - inhibit bladder and excite the bladder base and urethra
- Parasympathetic
 - S2-4
 - excite the bladder and relax the urethra
- Somatic
 - Onuf's nucleus
 - pudendal nerve
 - excite the external sphincter





Guarding Reflex
(Storage)

Voiding Reflex
(Emptying)

Afferent Pathway	Efferent Pathways
<i>Urine Storage</i>	
Low-level vesical afferent activity (pelvic nerve)	External sphincter contraction (somatic nerves)
	Internal sphincter contraction (sympathetic nerves)
	Detrusor inhibition (sympathetic nerves)
	Ganglionic inhibition (sympathetic nerves)
	Sacral parasympathetic outflow inactive
<i>Micturition</i>	
High-level vesical afferent activity (pelvic nerve)	Inhibition of external sphincter activity
	Inhibition of sympathetic outflow
	Activation of parasympathetic outflow to the bladder
	Activation of parasympathetic outflow to the urethra

Storage phase

- Myelinated A δ sensory fibres send afferent signals as low level firing (vesicosympathetic 'storage' reflex) results in :
 - Sympathetic pathway +ve
 - Parasympathetic pathway -ve
 - Somatic pathway +ve
- Results in
 - Detrusor relaxation (β_3 -ad)
 - Internal urethral sphincter contraction(α_1 -ad)
 - External urethral sphincter contraction(nico)

Emptying phase

- Higher level of firing send signals to PMC, also “OK” signal from cerebral cortex
- On-off switch is ‘flipped, PMC send signals to
- Sympathetic -ve
- Parasympathetic +ve
- Somatic -ve
- Results in
 - Detrusor contraction (M_3)
 - Internal urethral sphincter relaxation
 - External urethral sphincter relaxation

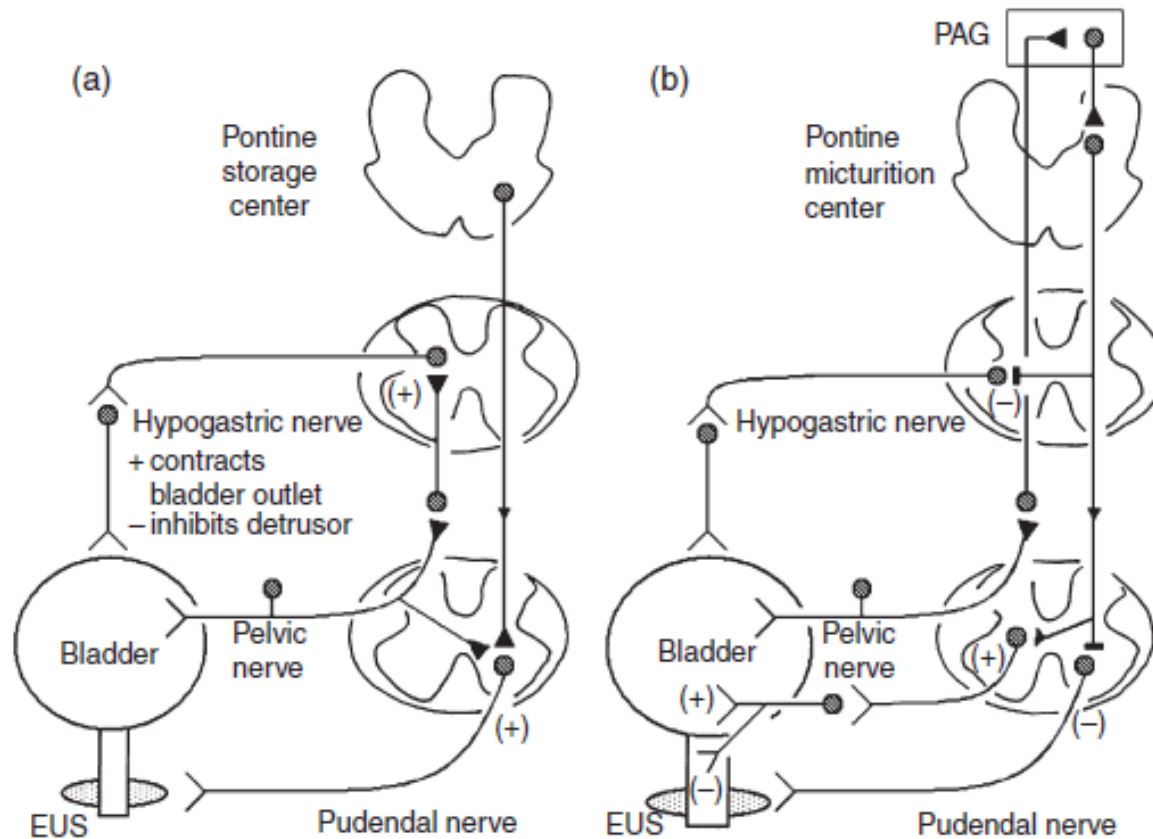
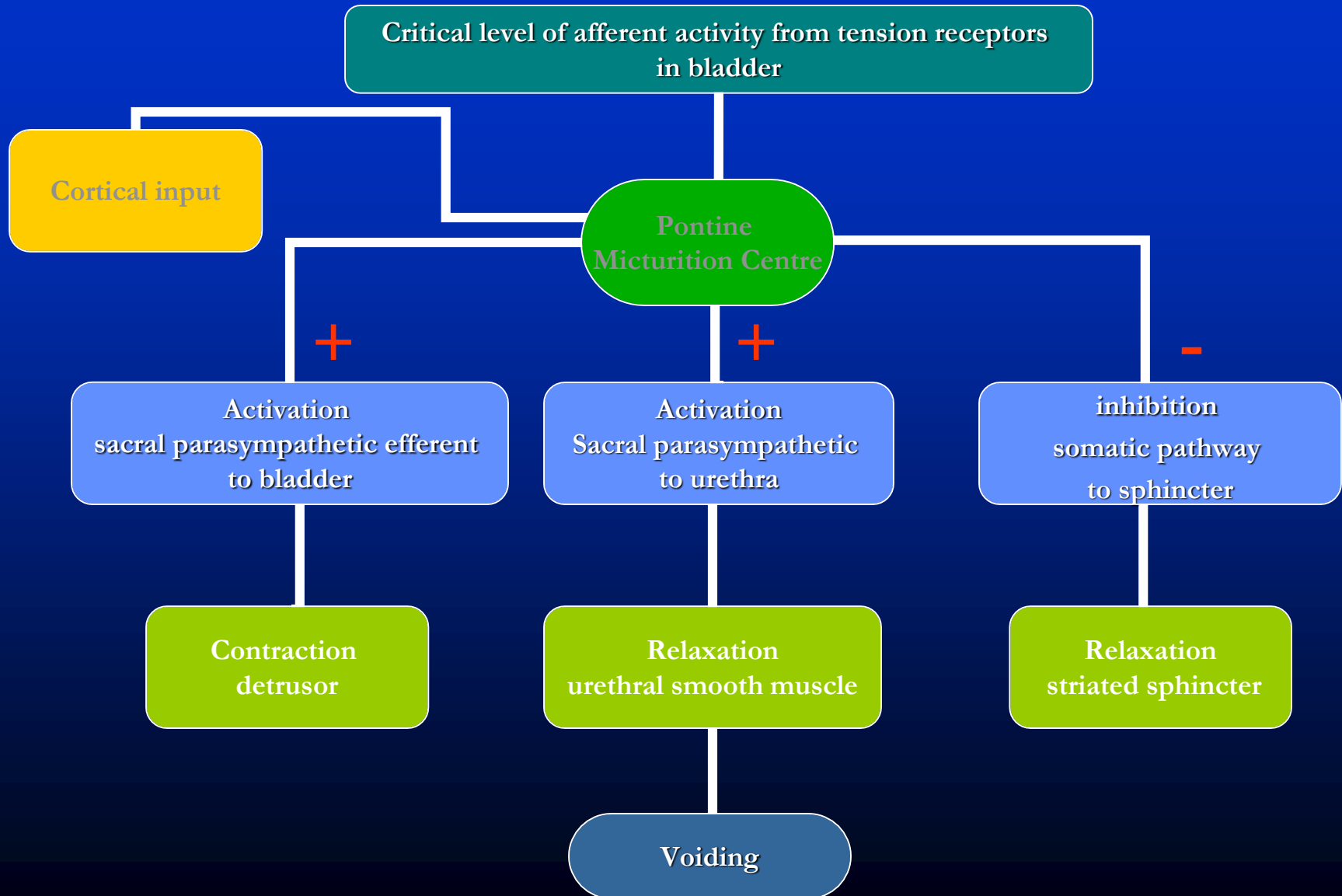
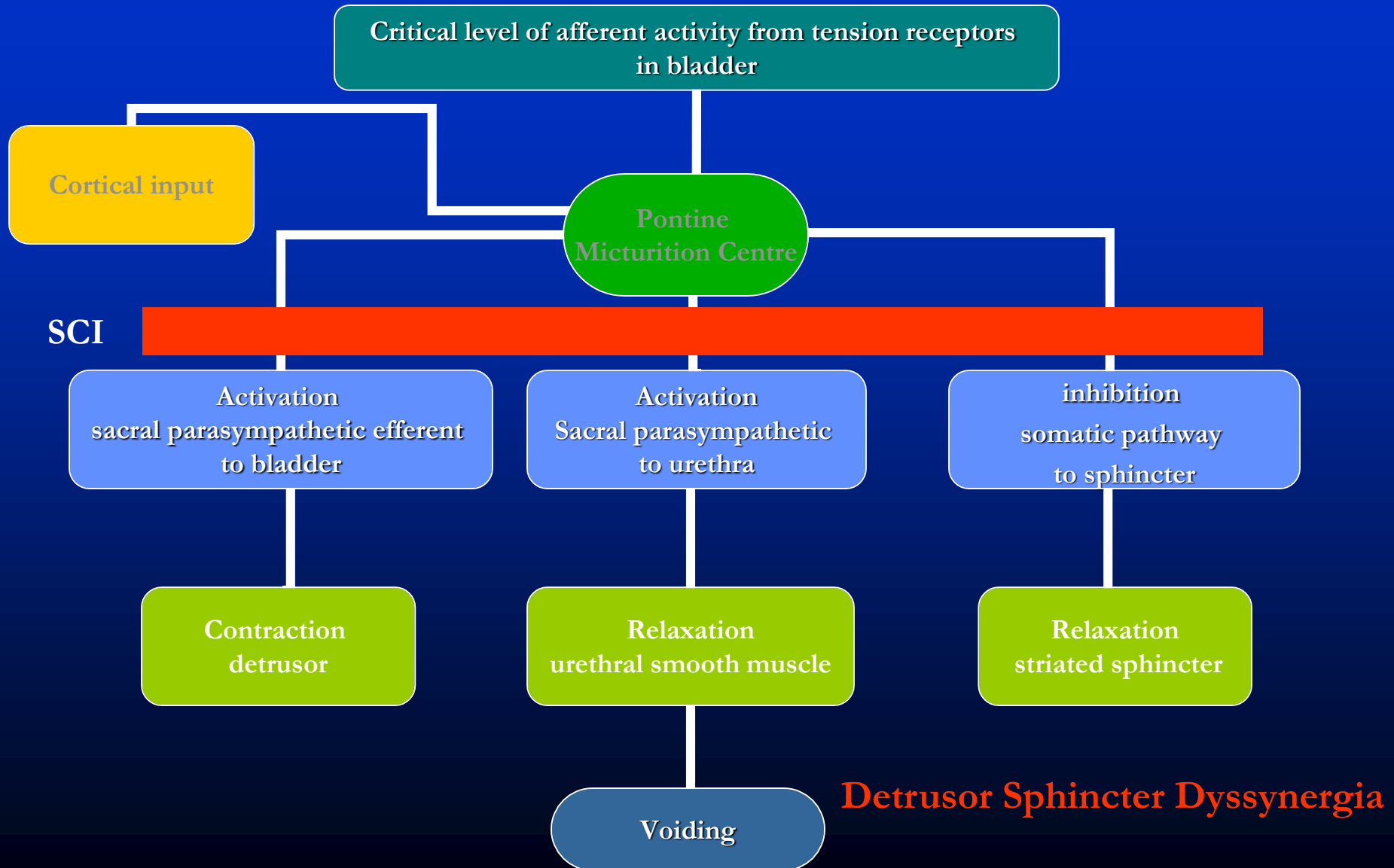


Figure 2.7 Diagram showing neural circuits controlling continence and micturition. (a) Urine storage reflexes. During the storage of urine, distention of the bladder produces low level vesical afferent firing, which in turn stimulates (1) the sympathetic outflow to the bladder outlet (base and urethra) and (2) pudendal outflow to the external urethral sphincter (EUS). These responses occur by spinal reflex pathways and represent guarding reflexes, which promote continence. Sympathetic firing also inhibits detrusor muscle and modulates transmission in bladder ganglia. A region in the rostral pons (the pontine storage center) increases external urethral sphincter activity. (b) Voiding reflexes. During elimination of urine, intense bladder afferent firing activates spinobulbospinal reflex pathways passing through the pontine micturition center, which stimulate the parasympathetic outflow to the bladder and internal sphincter smooth muscle and inhibit the sympathetic and pudendal outflow to the urethral outlet. Ascending afferent input from the spinal cord may pass through relay neurons in the periaqueductal gray (PAG) before reaching the pontine micturition center.

Voiding reflex



Voiding reflex



Pathophysiology of Urinary Retention



Retention of Urine

- Acute (AROU)
 - Sudden onset, painful
 - Eg. benign prostatic obstruction
- Chronic (CROU)
 - Usually painless, vague lower abdominal distension
 - Eg. Hypocontractile bladder

AROU / AUR

- Acute urinary retention
 - Common reason for hospital admission in Urology
 - Defined as “sudden inability to pass urine”
 - Can occur in both males and females

Incidence in men

- About 7 per 1000 men per year
- Overall cumulative probability of AUR in 4 years = 3%
- Over 5 yrs,
- 10% of men in 70s will have AUR
- ~30% of men in 80s will have AUR

Jacobsen SJ, Jacobson DJ, Girman CJ *et al.* Natural history of prostatism: risk factors for acute urinary retention. *J Urol* 1997; **158** : 481-7

Incidence in men (2)

- Among 100 men studied over 5 years
- Incidence of AUR ~2%

AJ Ball, P Abrams *et al*, BJU 1981

Risk factors (male AUR)

- Rates of AUR found to be increased with
 - Increasing age
 - Increasing prostate size
 - Increasing BPH symptoms
 - Decreasing maximal urine flow rate

Olmsted County Study

Incidence in women

- Not as well studied as in men
- Incidence ~3 / 100,000 person per year

Klarskov P *et al* Scand J Urol Nephrol 1987; 21(1): 29–31

- Bladder outlet obstruction less common
- A proportion has detrusor underactivity (DUA)

AROU : pathophysiology

- Mechanisms
 - Increased resistance to urine flow
 - Mechanical : eg. urethral stricture, clots, stone
 - Dynamic : eg. enlarged prostate / bladder neck tone
 - Functional : eg. neuropathic bladder
 - Impaired detrusor contractions
 - Drug induced : eg. antimuscarinics
 - Bladder overdistension : eg. post-anesthesia
 - Neuropathic bladder
 - Combination

Acute Retention Of Urine

- Common obstructive causes in males
 - Benign Prostatic Hyperplasia (BPH)
 - Cancer of prostate
 - Bladder/ Urethral stone
 - Bladder neck stenosis (usually after previous prostate surgery)
 - Urethral stricture (iatrogenic or infection/ inflammation)
 - Phimosis
 - Bladder tumour
 - Clot retention (Severe gross hematuria)

Table 2 Aetiology of AUR [16]

Benign prostatic hyperplasia	53%
Constipation	7.5%
Carcinoma of the prostate	7%
Urethral stricture	3.5%
Clot retention	3%
Neurological disorders	2%
Post-operative	2%
Calculus	2%
Drugs	2%
Infection	2%
Miscellaneous/unknown	16%

Murray K, Massey A, Feneley RC (1984)

Acute urinary retention—a urodynamic assessment. Br J Urol 56(5):468–473

TABLE 1 Causes of AUR in 170 men with LUTS in Europe, Latin America, Middle-East, Asia, and Canada [14]

Cause	%
BPH (spontaneous AUR)	64.8
Postoperative	13.1
Excessive fluid intake (alcohol)	12.5
Drugs	2.8
UTI	2.3
Urolithiasis	1.7
Acute medical condition	1.7
Post-traumatic	1.1

Acute Retention Of Urine

- Common causes in females
 - Detrusor hypocontractility - exclude DM
 - Neurogenic bladder
 - Idiopathic
- Neurogenic bladder
 - Bladder dysfunction associated with other neurological deficit
 - Eg. SCI, CVA, parkinsonism

Table 1. The common causes of female AUR

Anatomical	Organ prolapse e.g., cystocoele Mass effect Gynaecological tumour Urological tumour Constipation Bladder neck stenosis
Drugs	Opiates Antimuscarinics Antipsychotics α adrenergic agonists
Operative	Pain Anaesthetic related e.g., epidural anaesthesia Incontinence surgery
Neurological	Cord trauma Degenerative neurological disease e.g., Multiple sclerosis Cauda equine syndrome Fowler's syndrome
Infective	Urinary tract infection Genital herpes

Common precipitating factors

- Constipation
- UTI
- Anesthesia / analgesia
- Immobility
- Painful peri-anal conditions e.g. thrombosed hemorrhoids, perianal abscess
- Excessive fluid intake (esp alcohol)
- Drugs
 - With anticholinergic S/E
 - Antipsychotic, antispasmodics, H₁ antihistamines, antidepressants, ipratromium, disopyramide
 - With sympathomimetic S/E
 - Cold meds, MDMA

NOT anuria or oliguria

- Causes of NO urine production
 - Pre-renal ARF
 - Dehydration
 - Shock
 - Renal ARF
 - Acute renal failure e.g. Drug-induced, ATN
 - Nephritis

History 1

- Confirm urinary retention
 - Check first catheterized urine volume
- Painful vs Painless retention of urine
- First episode or Recurrent episodes
- Preceding Lower Urinary Tract Symptoms (LUTS)
- Other BPH complications
 - Gross hematuria, UTI (fever/ dysuria)
 - Bladder/ urethral stone (strangury)
 - Renal impairment (uremic symptoms)

姓名：

日期：

	無	少於五份一時間	少於一半時間	大約一半時間	多於一半時間	幾乎每一次	你的分數
1. 排尿不清 在過去一個月中，你是否經常有未能把尿排盡的感覺？	0	1	2	3	4	5	
2. 尿頻 在過去一個月中，你是否經常在排尿後兩個小時內又要小便？	0	1	2	3	4	5	
3. 排尿斷續 在過去一個月中，你是否經常在排尿時尿流斷斷續續？	0	1	2	3	4	5	
4. 尿急 在過去一個月中，你是否經常感到“忍尿”有困難？	0	1	2	3	4	5	
5. 尿流無力 在過去一個月中，你是否經常有尿流細弱的症狀？	0	1	2	3	4	5	
6. 排尿費力 在過去一個月中，你是否經常需要用力才能開始排尿？	0	1	2	3	4	5	

	無	一次	兩次	三次	四次	多於五次	你的分數
7. 夜尿次數 在過去一個月中，你晚上醒來小便的次數是：	0	1	2	3	4	5	

IPSS 總評分

總評分結果：0-7 分為輕度徵狀；8-19 分為中度徵狀；20-35 分為重度徵狀。

就排尿徵狀作生活質素評分	非常好	好	滿意	尚可	不滿意	不愉快	很痛苦
假如按現在排尿情況，你覺得今後的生活質素如何？	0	1	2	3	4	5	6

History 2

- Precipitating factors of AROU
- **Lower limb weakness/ numbness**
- **Overflow incontinence/ faecal incontinence**
 - Suspect ACUTE SPINAL CORD COMPRESSION
- Constitutional symptoms/ multiple bone pain
 - Suspect metastatic ca prostate

History 3

- Medications
 - BPH medications e.g. alpha blockers/ 5 alpha reductase inhibitors
 - Anticholinergics etc
- Previous history of prostate surgery e.g. TURP or urethral instrumentation
- History of STD especially gonococcal/ non-gonococcal urethritis
- History of CVA/ Parkinsonism/ Spinal surgery

Physical Examination 1

- Vital Signs and General exam.
 - Is the patient septic ?
 - Is the patient uremic ?
- Abdominal exam.
 - Palpable urinary bladder (if not yet catheterized)
 - Any phimosis
- Urethral catheter output - any gross hematuria?

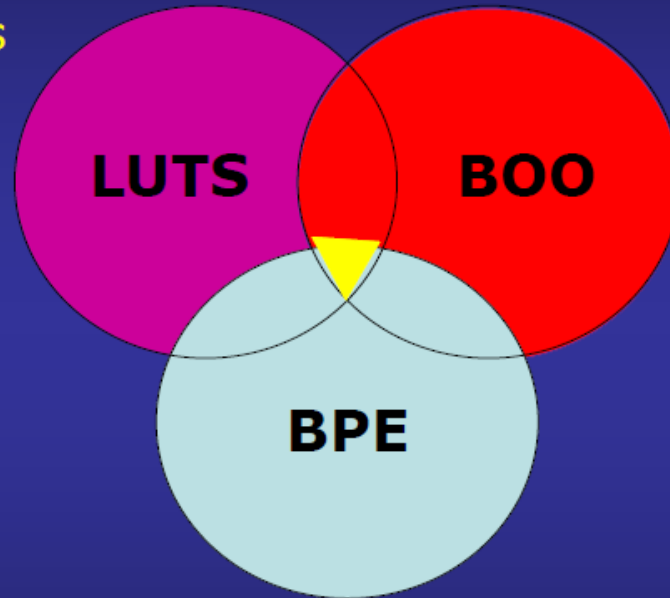
Physical Examination 2

- Digital Rectal Examination (DRE)
 - **Anal tone and peri-anal sensation**
 - Prostate estimated size
 - Any clinical features **suspicious of prostate cancer**
 - Prostate tenderness
 - Faecal impaction
 - Any painful peri-anal conditions
- **Lower limb neurological deficit**

Hald Diagram

Lower Urinary Tract
Symptoms

Bladder Outflow
Obstruction

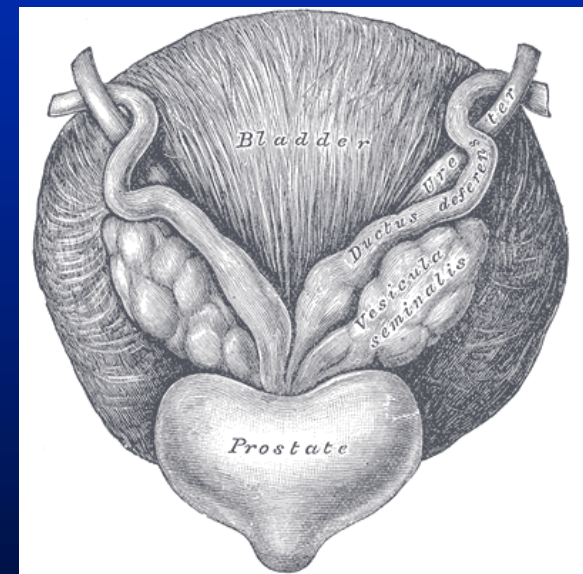


Benign Prostatic Enlargement

*Hald T et al. 4th International Consultation
on Benign Prostatic Hyperplasia,
Paris, 1997. SCI, 1993, 129-178.*

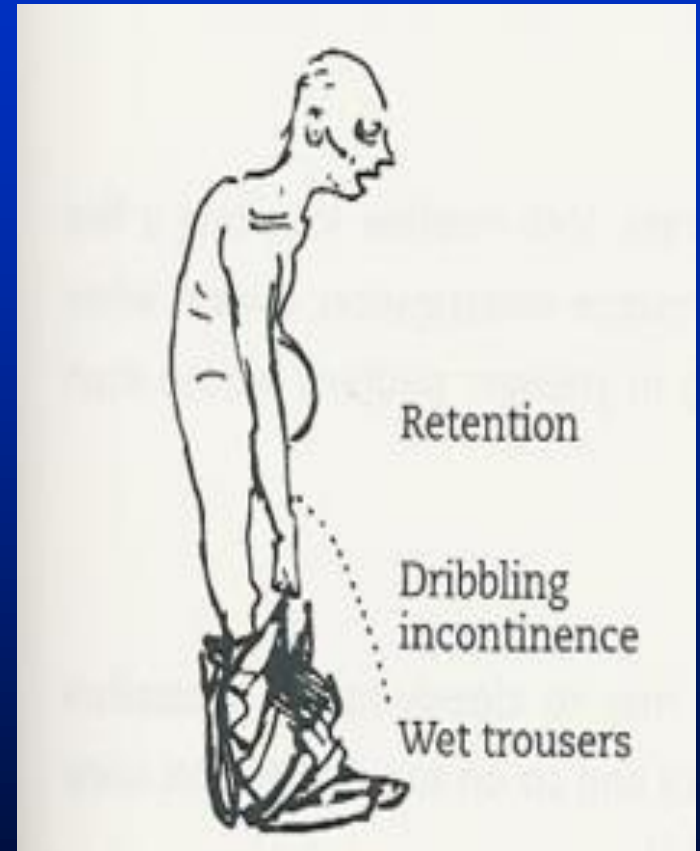
Bladder outlet obstruction

- More common in man
 - Male
 - Anatomical
 - Benign prostate hyperplasia (BPH)
 - Urethral stricture
 - Stone
 - Tumour
 - Functional
 - Sphincter dyssynergia
 - Female
 - Post operation e.g. after surgery for stress incontinence



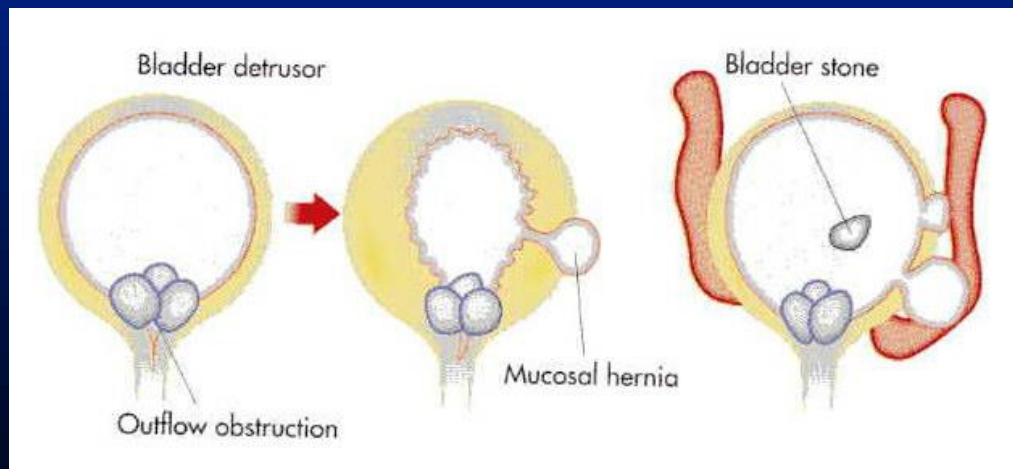
Symptoms of BOO

- Weak stream
- Hesitancy
- Intermittency
- Terminal dribbling
- Incomplete emptying
- Overflow incontinence
- Lower abdominal distension
- Uremic symptoms

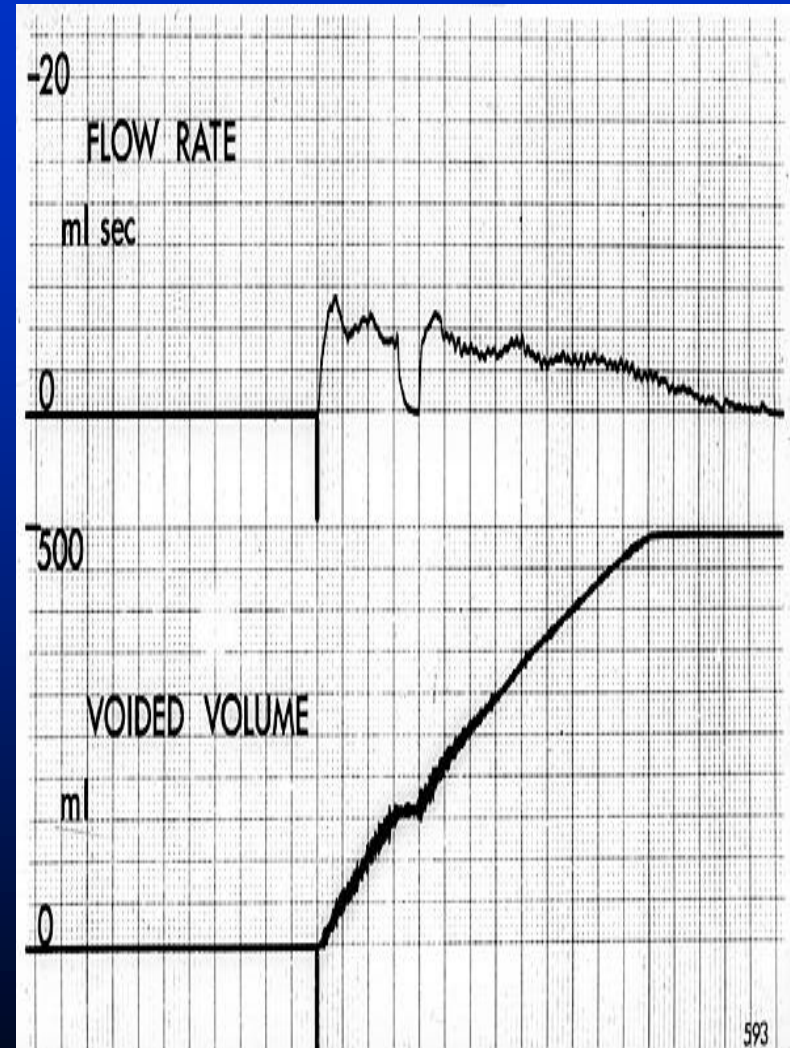
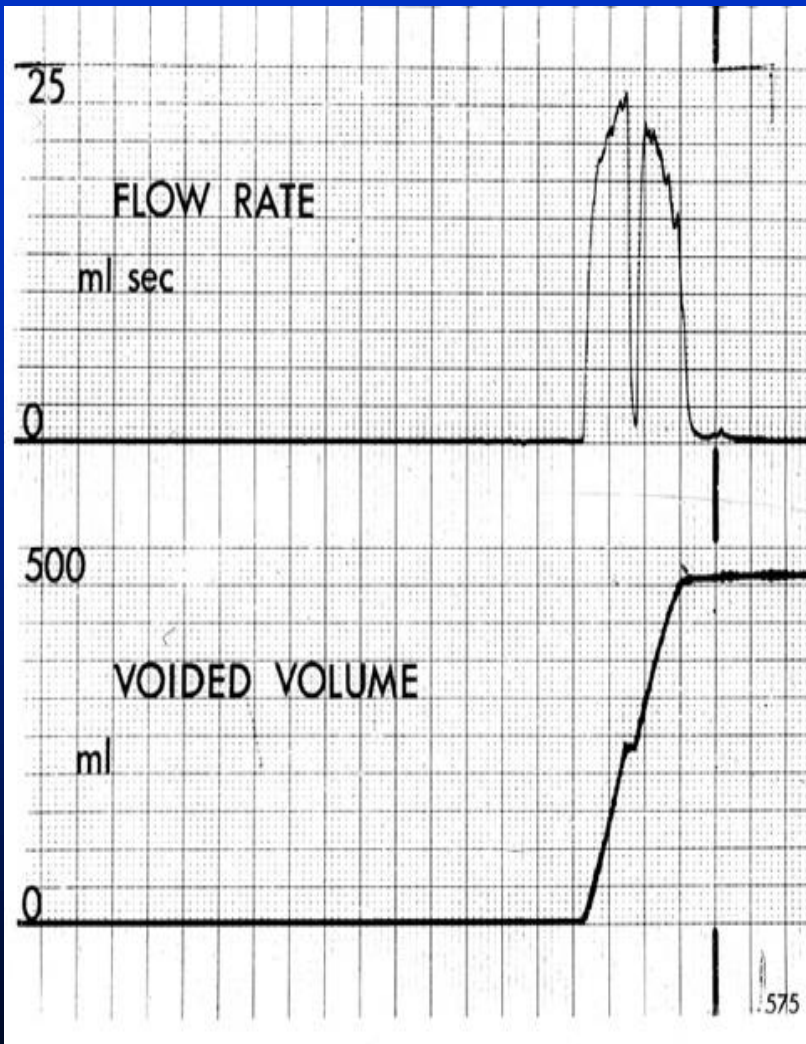


Consequence of BOO

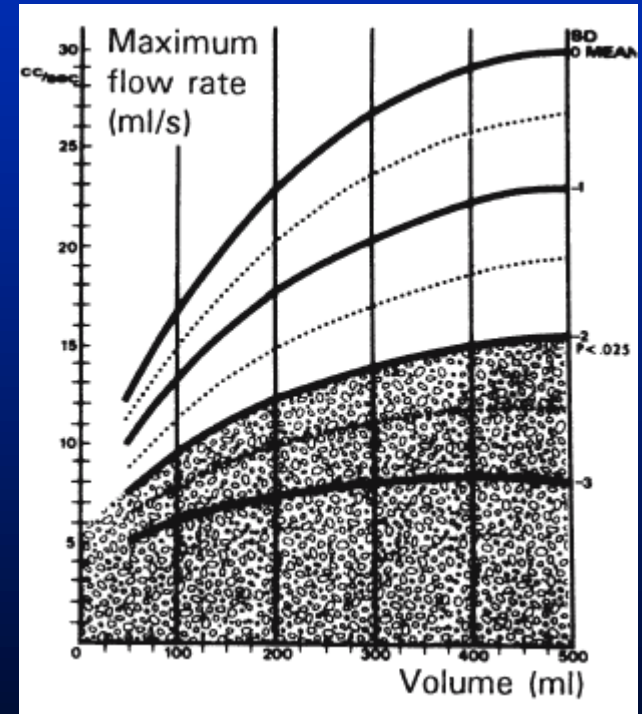
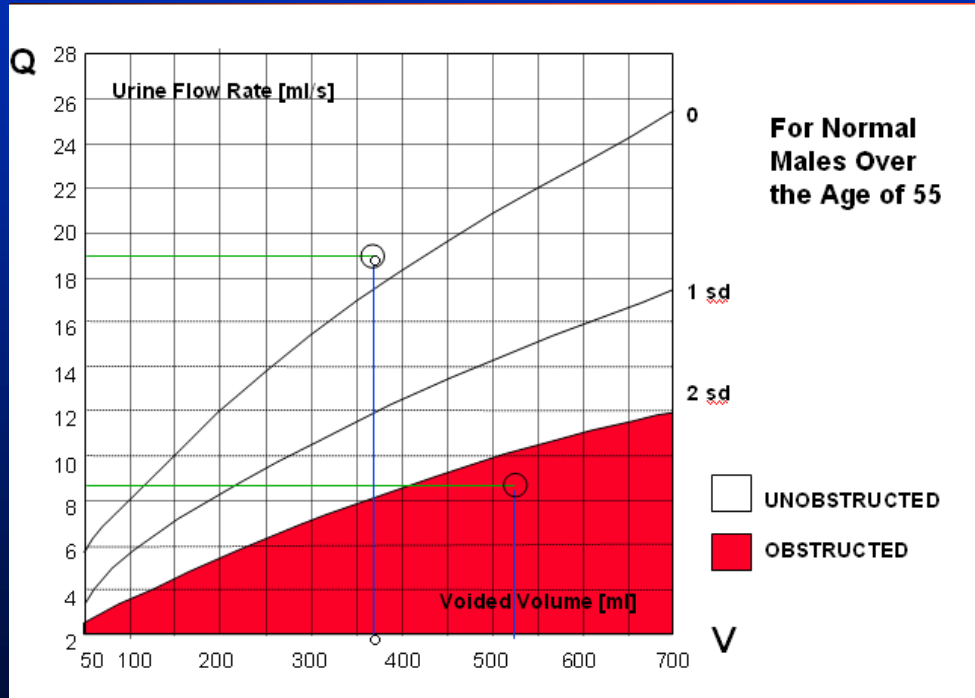
- Retention of urine - acute or chronic
- Recurrent UTI
- Formation of bladder calculi
- Hydroureter and hydronephrosis
- Renal impairment / ARF (Obstructive uropathy)



Uroflowmetry



Flow rate nomograms



Obstructed voiding = ?BOO

- Uroflowmetry NOT sufficient to diagnose outlet obstruction
 - Cannot distinguish obstruction from poor detrusor contractility
- Maximum flow rate (Q_{max})
 - Q_{max} 15-20ml/sec “normal”
 - Q_{max} < 10ml/sec “abnormal”
- 47% of BOO pts have <10mL/s (PPV 70%)
82% of BOO pts have <15mL/s (PPV 67%)
- **18% are obstructed despite $Q_{max} > 15$**
 - Reynard JM et al. “The ICS-‘BPH’ Study: uroflowmetry, lower urinary tract symptoms and bladder outlet obstruction.” British Journal of Urology. 1998; 82: 610-623.

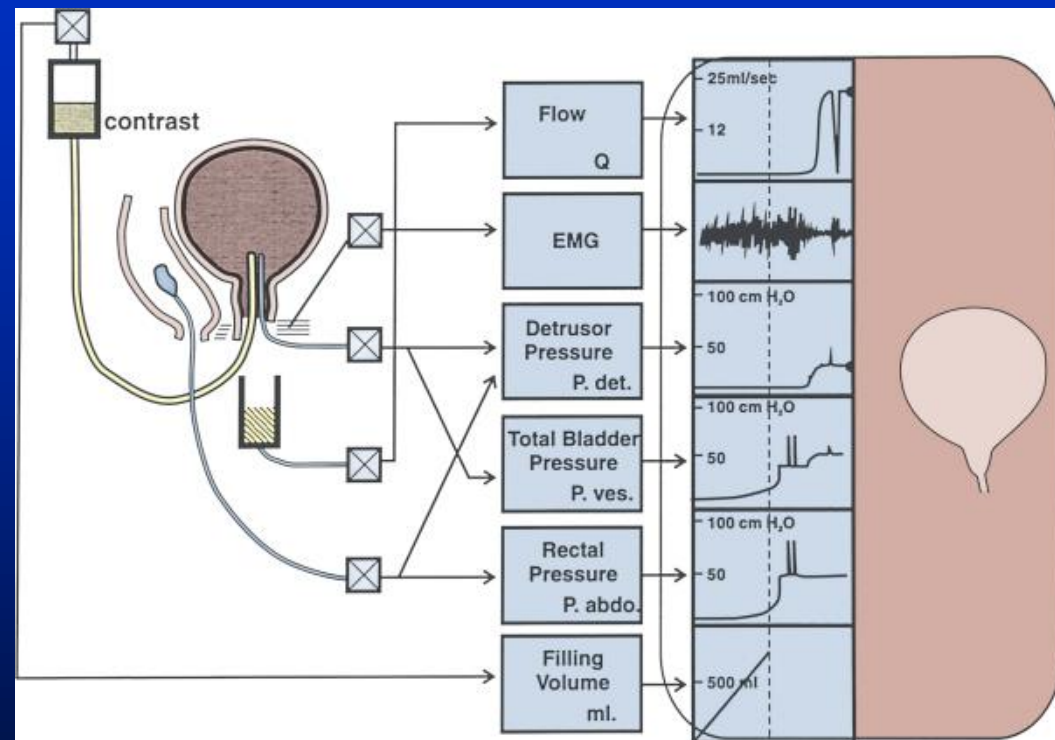
Table 3.1. Lowest acceptable maximum urine flow rates according to age and sex for minimum voided volumes^a

Age years	Minimum Volume ml	Male ml/s	Female ml/s
4-7	100	10	10
8-13	100	12	15
14-45	200	21	18
46-65	200	12	15
66-80	200	9	10

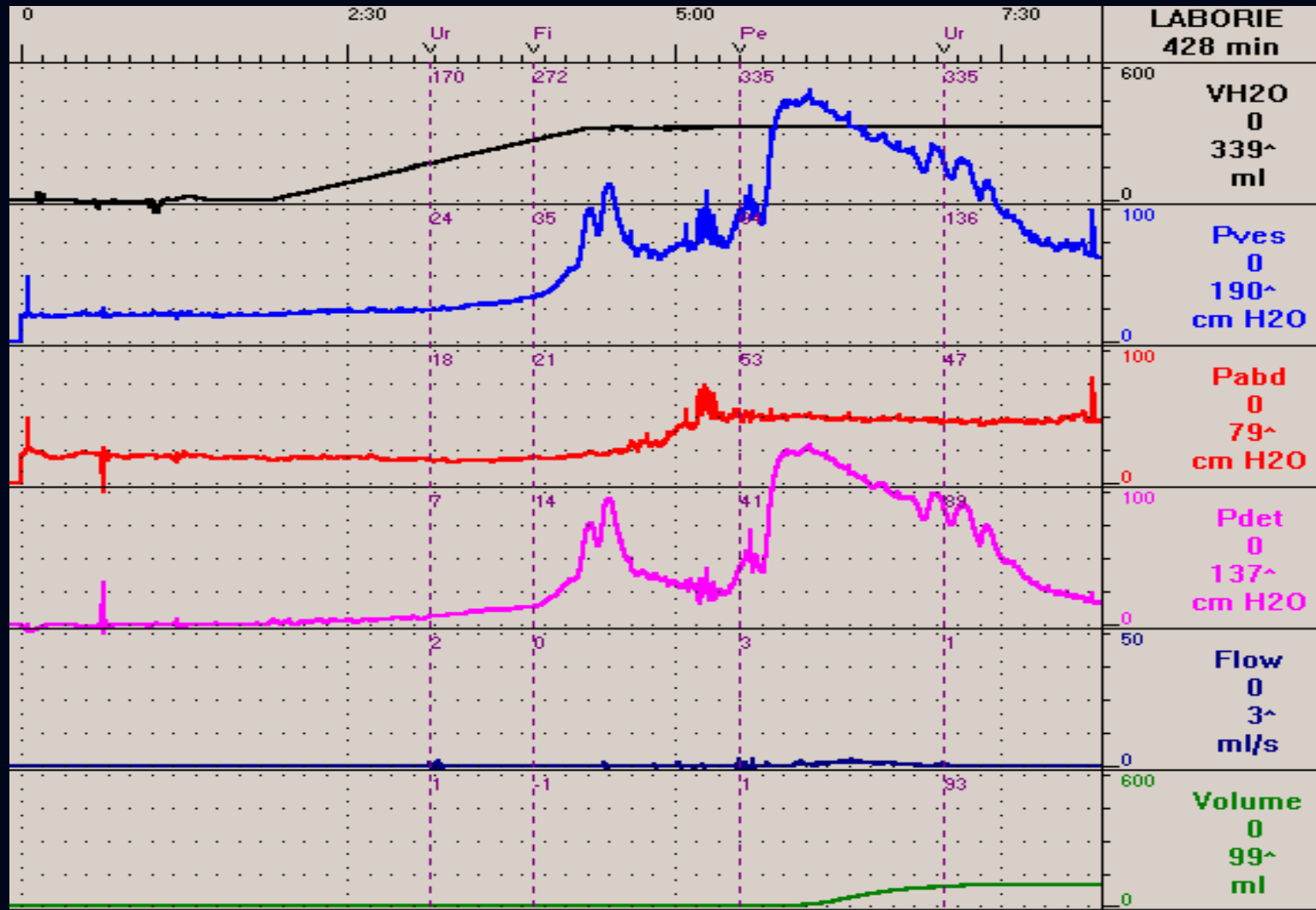
^aValues given are taken from personal experience and relevant literature. In general the values are one standard deviation below the mean for the maximum flow. Values below those given may not be abnormal, but need further consideration.

Urodynamic studies

- Study function of the lower urinary tract
- Parameters measured
 - Intravesical and rectal pressure
 - Detrusor (deduced)
 - Uroflow rate
 - Sphincter function EMG
 - Bladder volume
- Cystogram and reflux (video)

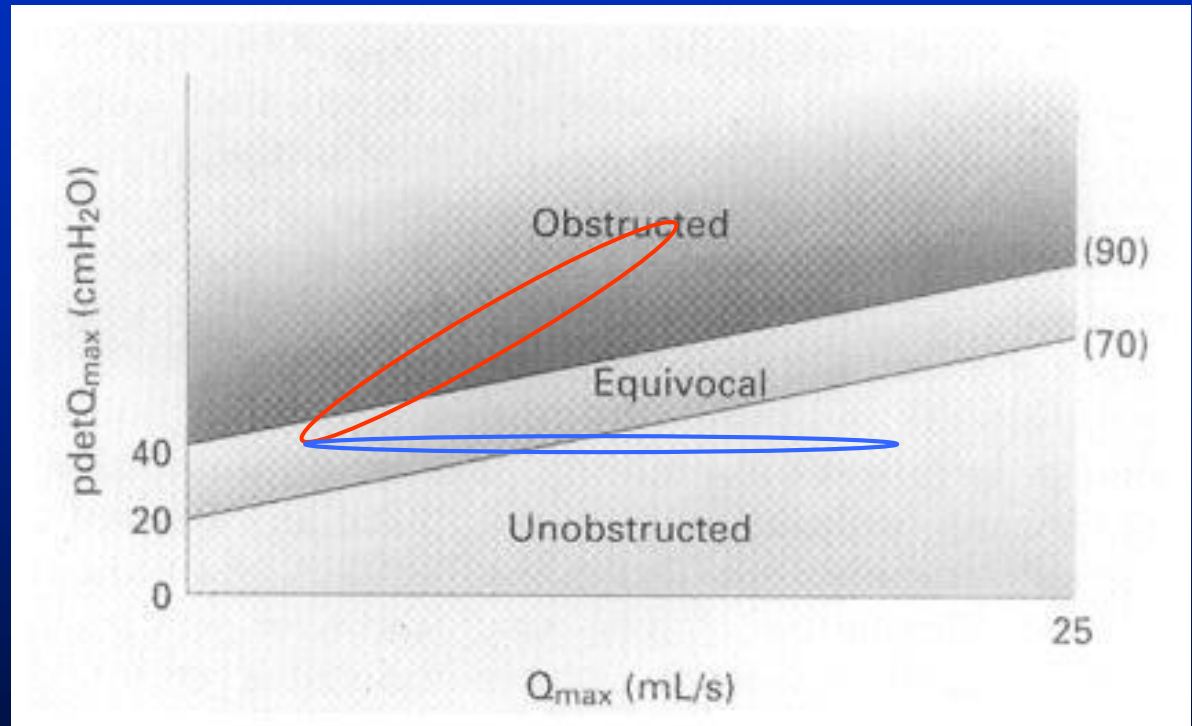


Bladder outlet obstruction



Bladder outlet obstruction

- Pressure flow study “Low uroflow rate with high detrusor pressure”
- BOO =
Urodynamic
diagnosis



Clinical Management of Urinary Retention



Management 1

- Immediate decompression of bladder
 - Urethral catheterization
 - Suprapubic catheterization (SPC)
- SPC indicated in
 - Failed urethral catheterization
 - History of urethral trauma eg. straddle injury

Management 2

- SPC contraindicated in
 - Non-distended bladder
 - Uncorrected bleeding tendency
 - Known / suspected urothelial cancer
- SPC complications
 - Bowel perforation
 - Rectal injury (overshoot)
 - Hematuria

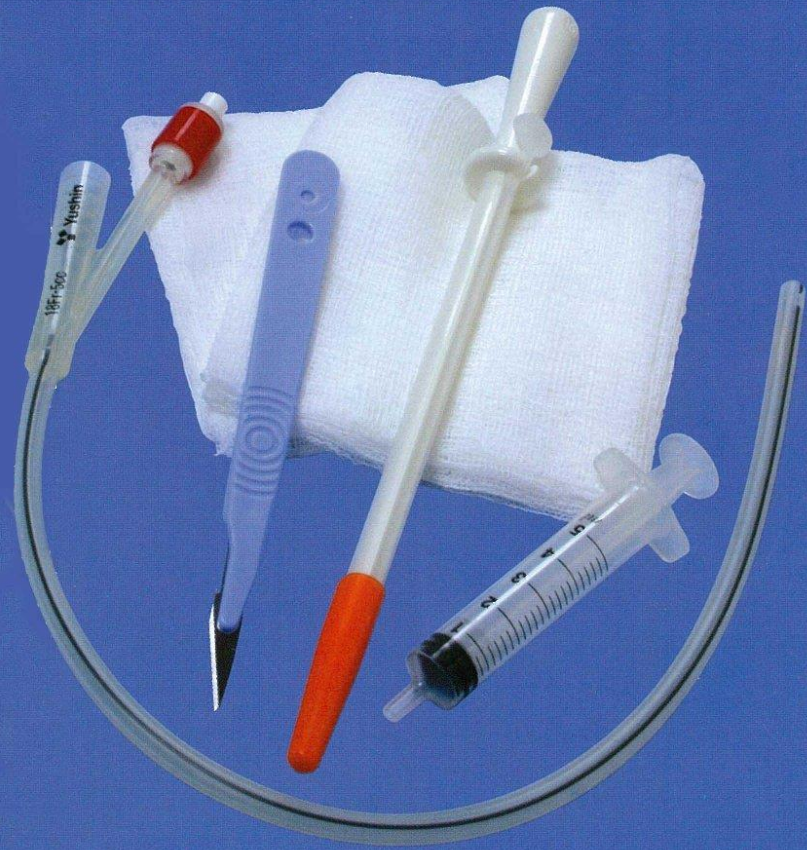
Indications

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Introduction

1





Investigations

- CBP, L/RFT
- CSU x C/ST
- KUB

- No PSA
- Likely falsely elevated

Learning objectives

- Learn the pathophysiology of micturition
 - Structure and function
 - Storage vs emptying phase
- Understand how urinary retention develops
- Grasp the concept of BOO
- Discuss the clinical management of urinary retention



